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SDE-2

PROFILE

Software Development Engineer II on Groupon's **Pricing & Promotions** platform: **Incentive Service** (personalized discounting, promo codes, invite-only deals), **Message Service** (push notifications, emails, web banners), and **Pricing Service** (flash sales, dynamic pricing, pricing models). Delivers high-throughput, cost-efficient, reliable backends (**Java/Spring**, **GCP/Kubernetes**, **Kafka**, **Redis**, **Postgres/Bigtable**) with strengths in p99 tail-latency reduction, incident response (QRT) and practical LLM/AI (search re-ranking, email subject/body copy, send-time optimization).

SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, C/C++

Backend & Frameworks: Spring Boot, Play, Akka, Node.js, gRPC, WebSockets, Next.js

Data & Messaging: Postgres, Bigtable, MySQL, Redis, MongoDB, Kafka, ELK

Cloud & Infra: GCP, AWS, Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitHub Actions)

Observability: Grafana (Wavefront migration), Prometheus, structured logging, alerting, runbooks

Testing: JUnit, Testcontainers, Pact; performance profiling; reliability hardening

Domain: Pricing & Promotions, promo codes, personalization, notifications, dynamic pricing

EXPERIENCE

Groupon

Oct 2024 – Present | Software Development Engineer II

▪ Cost & Efficiency

- Led multi-service rightsizing (Conveyor Cloud) with SLO guardrails; built CPU/memory utilization curves, recommended new limits, coordinated safe rollouts, and verified post-change metrics → **~\$50k/yr infra savings**.
- Downsized Bigtable capacity and backup footprint with metric watches → **~\$70k/yr savings** and a leaner backup strategy.
- Eliminated batch OOMs by heap profiling and batch-size tuning; **memory 16 GB→8 GB**, stable peak runs with zero regressions.
- Reduced noisy logs across MS/IS; faster triage and lower log storage costs.

▪ Performance & Reliability

- Targeted worst tail-latency endpoints, partitioned workloads, added retries, and shipped dashboards → **errors ↓90%, avg CPU ↓50%, latency peaks ↓75%**.
- Uplifted audience-load throughput with "Current Queue" visibility, EMEA tuning, and better handling of rejected audiences → **2× speed, ~\$10k savings**.
- Optimized MDS deal-load payloads (smaller payloads, better pagination, tuned timeouts) → **timeouts reduced to zero**.
- Hardened min-price cache with early-warning alerts → **fewer breach investigations**.

▪ Growth & Revenue

- Built customer-facing **promo-code visibility** (Next.js) with category mapping and weekly themes → **~\$500k M1VFM**, consistent messaging.
- Shipped **giftable deal banners** under holiday deadlines; simplified marketer-led content → **~\$600k seasonal impact**.
- Built a **checkout upsell system** suggesting better deals on higher spend/quantity → **+14% revenue, +20% order size, +10% conversion, +70% more multi-item orders**.

▪ Automation & Ops Enablement

- Implemented **recurring IS campaigns** with audit trails → **lower ops toil, on-time execution**.
- Delivered **Bulk Create** (CSV → validated audiences + taxonomy + MDS search) → **576 campaigns in 23 days, ~\$50k ops savings**.
- Proactive **IS alerts** for starts <24h and long expiries → **~\$300k risk mitigation**.
- **Promo retention post-auth failure**: preserved eligibility across retries → **~12% WoW fewer refunded duplicates**.
- 15-minute **feeds refresh** to propagate purple-price changes → **fresher storefronts, fewer manual fixes**.
- **ILS tagging automation**: attribute-driven tagging; eliminated manual effort, improved discoverability.

▪ Cross Team Collaboration

- **Bloomreach push parity**: validated payload schema/back-compat; reduced integration risk and enabled pilot scope.
- **CDP logging + Airflow jobs**: migrated email headers to Hive; stood up proxy logging and validated downstream analytics.
- **AI personalization pilots** (Spark/Airflow): LLM subject lines and deal titles → **titles negative CTR; subjects flat OR with positive CVR/Rev spikes (low-N)**; shipped learnings with no cost/perf regressions.
- **MCP server for deal search** (Spring MCP + LLM re-ranking): NL query → structured filters (category/geo/price/tags); modular backend so web/app search can opt-in without UI rewrites.
- **AI for marketing emails**: LLM-assisted subject/body suggestions and send-time heuristics with safety guardrails, offline eval harness, and human-in-the-loop via CRM-Message-Service.

▪ Peak-Week Readiness & QRT

- Ran a **24×7 command center** across revenue/traffic, discount end-time tuning, and platform QA; triaged live issues (purple-price carousel, promo translation, coupon rec formatting) → **protected GP and CX under load**.
- Core **Quick Response Team (QRT)** responder for pricing, incentives, and CRM incidents; restored services, led post-mortems, and produced runbooks/dashboards to prevent regressions.
- Reliability hardening: cache freshness, alert revamps, batch retries, static analysis fixes, and log analysis → **faster detection/recovery**.

Groupoon

Jul 2023 – Sep 2024 | Software Development Engineer I

- Migrated core services **AWS→GCP** → ~ 40% performance uplift and ~ \$500k/yr savings; built multithreaded data pipelines to move critical tables in minutes.
- Implemented real-time **affiliate incentive qualification** with idempotent flows and seamless redemption UX.
- Enforced **merchant caps** across **44k+** deals for fairness and abuse prevention.
- Shipped **CSV-driven campaign creation** consolidating audience validation, taxonomy, and MDS search → **576 campaigns/23 days**.
- Enhanced **CRM-Message-Service** for banners/emails/push to improve consistency and deliverability.

POSITION OF RESPONSIBILITY

Web System Administrator, NIT Allahabad

Jan 2021 – 2023

- Operated the college web server and academic portal; maintained digital records, grade submissions, hostel allotments, and admissions.
- *PHP, Django, MySQL, Linux (CentOS), Apache*

PROJECTS

AmbuFast — Node.js, React, MongoDB, Socket.io, Mapbox

Hackathon Top-5 (2023)

- Real-time ambulance tracking with SOS; route broadcast to traffic control; doctor/patient dashboard with Aadhar-linked history.

Checker's King — MERN, TailwindCSS, WebSockets

GitHub

- Online multiplayer checkers with video/chat, ELO ratings, spectator mode, and a real-time WebSocket engine.

EDUCATION

B.Tech, Computer Science & Engineering — NIT Allahabad

Jul 2019 – Jul 2023

GPA: **8.59/10**

Class XII (CBSE Board)

2019

Percentage: **95.2%**

Class X (CBSE Board)

2017

CGPA: **10/10**

SELECTED ACHIEVEMENTS

- **Star Performer**, Groupoon — recognized for raising the bar and going the extra mile.
- **Top 10**, Hack36 (36-hour hackathon) among 100+ teams.
- **Winner**, Webster (Avishkar 2021); **1st Runner Up**, Reveng CTF (Avishkar 2021).

ADDITIONAL INFORMATION

Core competencies: Distributed systems, system design, performance engineering, promotions/personalization, cost optimization, reliability/observability, QRT

Tools: GitHub, Perforce, ELK, Grafana, Kafka, CI/CD, code reviews

Interests: Platform reliability, data-driven growth, clean architecture, mentoring